

OpticNeuroBlender: auditable coherent scalar learning from evaluated Blender geometry

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Abstract

We implement a coherent scalar optical classifier whose propagation and trainable phases originate from evaluated Blender geometry. A rational geometric state graph preserves every nonzero admitted branch and merges only identical future optical states. An independently written auditor checks nearest intersections, branch coverage and local surface neighbourhoods. The captured demonstrator contains 104 admitted objects, five sources, eight terminal modes and 16 paired mirror-translation parameters. Its complete graph has 133 states, 186 edges and 17,060 terminal path occurrences. Independent outward-rounded rational field composition bounds the errors of observed outputs and permits decisions only when power intervals separate. Geometry-driven training achieves 27/30 held-out Iris predictions after 60 fixed updates; actual Blender save/reopen inference reproduces all 150 predictions. On a prospectively fixed Wine split, three initializations obtain 33/37, 30/37 and 28/37, compared with 32/37 for matched-input linear and quadratic classifiers. Actual NVIDIA complex128 inference and gradients agree with CPU references, but small batches are slower and an equivalent dense baseline is more efficient. Separately, actual native graphics shaders select captured surfaces and compute coherent FP64 phase/fields; independent rational intervals bound all 450 observed outputs and certify all 150 unique decisions. Native graphical derivatives for 16 geometric parameters pass 32 independently audited finite-difference geometries. An identical-proof CPU-BVH/GPU-raster ablation observes no full GPU advantage in the captured scene. These graphical computations preserve an explicit CPU topology, affine-tangent, coherent-merge and readout boundary. A declared Gaussian scalar wave reference exposes a finite-window failure, retained before a separately frozen remedial study. A separately frozen GitHub-hosted Linux CPU run reproduces 150/150 decisions and 27/30 held-out predictions, with maximum power difference 3.89×10^{-16} . A separate clean native Linux Blender run passes nine installation/training/recovery controls. Physical-network fidelity, AMD execution, exceptional novelty and independent expert replication remain unestablished.

1. Question and falsifiable scope

The operational question is whether an accessible scene editor can serve as a reproducible instrument for an own optical neural model, with traceable geometry, coherent fields, learning and useful numerical certificates. The represented model is ideal, scalar, monochromatic, lossless planar optics in a homogeneous medium. It is not a calibrated photonic processor or a complete Maxwell model.

The certificate claim is falsifiable: one independently verified exact represented-model output outside a claimed enclosure, one invalid certified decision, or one unbounded omitted nonzero admitted branch refutes that claim. Completeness, geometric correctness, enclosure correctness and classifier accuracy are separate acceptance conditions. A solver that rejects every useful input does not demonstrate utility. A timeout is not a zero optical field or a scientific success.

The original 90-second pilot was executed without extending its deadline and remains inconclusive. The exact indexed variant also timed out. Subsequent finite-state graph, training, GPU and wave protocols were distinct, frozen and published before execution under explicit human GitHub authorization. External registration/IPFS remains an additional pending route; no external identifier is claimed, and a later timestamp would not retrospectively preregister earlier observations.

2. Model, units and representation

2.1 Evaluated scene ingress

The capture reads Blender’s evaluated dependency graph: represented mesh vertices, triangle indices, instance transforms, object identities, resolved input/parameter/detector bindings and scene unit metadata. Hexadecimal floating-point values preserve exposed numbers. Rational arithmetic constructs world geometry from those represented values. This defines an exact affine represented model; it does not certify the unobserved geometry intended by an author or reproduce every rounding operation in Blender’s native transform/render pipeline.

The current scene admits 104 optical objects and 6,656 triangles, five sources, eight terminal modes and 32 coordinate bindings, grouped into 16 pairs of mirrors. Coordinates and wavelength use Blender units (BU); λ is the represented value of 0.1 BU. Scene display scale is not measured physical calibration. Reported powers are normalized scalar modal powers, not calibrated watts.

2.2 Coherence and detector semantics

$$E_p(\mathbf{x}, \boldsymbol{\theta}) = \sum_s x_s \sum_{\gamma: s \rightarrow p} A_\gamma(\boldsymbol{\theta}) \exp\left(i \frac{2\pi L_\gamma(\boldsymbol{\theta})}{\lambda}\right), \quad (1)$$

$$P_p = |E_p|^2, \quad e^{-i\omega t} \text{ temporal convention.} \quad (2)$$

Inputs are explicit complex amplitudes in one common launch-phase reference. The temporal convention is $\exp(-i\omega t)$; propagation contributes $\exp(+i2\pi L/\lambda)$. Declared mirrors and splitters use explicit phase/sign conventions. Summing intensities from independently labelled sources would remove cross terms and implement a different classifier. Fluence alone does not determine the complex field.

For a terminal mode, the field is the coherent sum of its admitted path contributions and power is its squared magnitude. Geometrical mode projection and completeness precede readout. The three class detectors are fixed. An estimated argmax is not automatically certified: certification requires the winning lower power bound to exceed every competing upper bound. Zero input and overlapping bounds retain an unknown decision.

2.3 Complete finite-state propagation

A state identifies represented origin, direction and previous optical plane. Two states merge only when these quantities are exactly identical, so their future propagation operator is identical. Every nonzero transmission/reflection remains an edge. Only mathematically exact zero coefficients omit

an edge. Cycles, unresolved intersections, missing support or resource limits produce incomplete status with no fabricated field.

An exact conservative object-AABB index narrows candidate work without changing the closed triangle queries. Shared triangle vertices do not become artificial optical edges: an independent projected square-perimeter witness checks an interior neighbourhood of the union of coincident triangles. Nearest-hit and branch auditing uses a separate implementation. The captured complete graph has 133 states and 186 edges while preserving 17,060 terminal path occurrences. This is a representation of the admitted scalar network, not a proof that a real optical device has only those physical paths.

2.4 Geometry-driven derivatives and training

$$\frac{\partial P_p}{\partial \theta_j} = 2 \operatorname{Re} \left(E_p^* \frac{\partial E_p}{\partial \theta_j} \right), \quad (3)$$

$$\mathcal{L} = -\frac{1}{N} \sum_{n=1}^N \log \frac{\exp(P_{n,y_n}/T)}{\sum_c \exp(P_{n,c}/T)}, \quad T > 0. \quad (4)$$

For fixed plane normals and admitted paired translations, ray/plane intersections yield exact affine origin, hit and segment-length derivatives. A merge is admitted only when its incoming affine origin derivatives agree. The phase derivative follows from the geometrically derived optical length. Own coherent forward derivatives propagate through every graph edge; power derivatives use $2 \operatorname{Re}(E^* \partial E / \partial d)$. No historical analytic weight matrix supplies the trained field or its gradients.

Training minimizes cross entropy of the three detector powers divided by a fixed loss temperature. Adam updates 16 mirror-pair world-X translations. Proposed positions are quantized to Blender’s represented native float32 coordinates and constrained around a declared untrained centre. Every evaluated optimizer state is independently audited; a fresh final geometric rebuild checks the affine computation. Finite differences and a separate functional autograd reference are software/numerical controls, not rigorous universal gradient certificates.

2.5 Capacity limitation

$$\mathbf{E} = U(\boldsymbol{\theta})\mathbf{x}, \quad (5)$$

$$P_c = \mathbf{x}^\top Q_c \mathbf{x}, \quad Q_c = \mathbf{a}_c \mathbf{a}_c^\top + \mathbf{b}_c \mathbf{b}_c^\top, \quad \operatorname{rank} Q_c \leq 2 \quad (\mathbf{x} \in \mathbb{R}^S), \quad (6)$$

$$\operatorname{rank}(Q_c - Q_d) \leq 4. \quad (7)$$

At fixed geometry, the network is linear in complex input: $\mathbf{E} = U\mathbf{x}$. Detection introduces a quadratic readout. For real input, one detector has $P = \mathbf{x}^\top Q \mathbf{x}$ with $Q = \mathbf{a}\mathbf{a}^\top + \mathbf{b}\mathbf{b}^\top$ when its field row is $\mathbf{a} + i\mathbf{b}$. Consequently Q is positive semidefinite with rank at most two; a difference between two such powers has rank at most four. For complex input, the corresponding Hermitian detector matrix has rank at most one. These algebraic restrictions hold independently of a numerical rank threshold.

The current model has no intermediate optical nonlinear activation and does not establish universal neural approximation. Normalizing the common input power is part of the declared encoding, shared with the classical comparisons. A dense matrix compiled from the graph’s five basis responses is an equivalent fixed-geometry baseline, useful for auditing and cost comparison.

A separately frozen independent five-basis interval composition of the final native trained graph encloses the full 8×5 operator. Hermitian interval LDL certifies column rank five. Gershgorin bounds enclose every eigenvalue of $U^\dagger U$ within $[1-\varepsilon, 1+\varepsilon]$, $\varepsilon \leq 3.0665868333669135 \times 10^{-19}$, proving $|||U\mathbf{x}||^2 - ||\mathbf{x}||^2| \leq \varepsilon ||\mathbf{x}||^2$ for every complex represented input $\mathbf{x}$. This fixed-geometry approximate

isometry includes all eight terminal modes; the three class detectors are only a subset. It is not an exact identity or calibrated physical-flux result. Midpoint eigensolver values are diagnostic only; a proposed vector is separately checked with rational Rayleigh intervals. Full analysis costs 6.468 s and 46.207 MiB RSS.

3. Independent numerical checks

$$P_c \in [\underline{P}_c, \overline{P}_c], \quad \underline{P}_c > \max_{d \neq c} \overline{P}_d \implies c \text{ is a certified argmax.}$$

Otherwise the decision remains unknown; classification correctness is a separate statement.

The independent output enclosure uses rational outward rounding at 128 bits, interval π from Machin series with remainder, square-root bounds from integer arithmetic and exact rational Taylor evaluation with explicit remainders. Complex sums and powers are composed through the audited graph. This bounds an already observed native output against the exact represented model rather than claiming that every libm or GPU execution has the same error.

The first enclosure was valid but too loose: field L1 bound 1.51×10^{-7} and power bound 1.93×10^{-7} exceeded the fixed 10^{-11} budgets. It remains archived. The revised exact-midpoint Taylor composition was published before the secondary recomputation, with the budgets unchanged. The final maximum bounds were $4.9383434427813286 \times 10^{-15}$ for complex-field L1 error and $3.1372973883238074 \times 10^{-15}$ for power error. The represented unit-input decision R1 had a lower margin of 0.18933601504039158. A separately written 90-digit full-graph composition fell inside every reported interval. The rational construction supplies inclusion; the high-precision computation is a cross-check.

An independent trained-batch certificate covers all 150 observed fields, powers and ideal train-only encodings: maximum field L1 bound 2.7683×10^{-15} and power bound 1.0366×10^{-15} . All decisions separate; 137 are correct and 13 wrong. Its input reference begins with parsed binary64 features, rather than unobserved measurement values.

These certificates do not fill missing budgets for intended geometry, native transform preimages, uncertain physical parameters, polarization, diffraction, detector coupling or material calibration. Those errors remain unknown. The unit-input decision certificate is not a certificate of all Iris or Wine predictions.

A separately frozen conditional robustness study defines 16 paired-translation boxes by half-adjacent binary32 coordinate spacing, expanded by factors 1, 10, 100 and 1000. Independently derived exact affine plane expressions prove selected surface-union support and exclude all 13,699 competitor-object opportunities across 133 states per box. All four continuous topologies are proved; interval coherent fields preserve 147/150, 131/150, 40/150 and 0/150 class argmax decisions respectively. Remaining decisions are unknown, not demonstrated unstable. At the smallest box (maximum shift 9.537×10^{-7} BU), 137 certified decisions are correct and ten are wrong. These are conditional represented perturbations, not verified full Blender transform/mesh preimages or manufacturing tolerances. Conservative field/power variation bounds are distinct from observed arithmetic error budgets.

A separate whole-family certificate proves the original training box (centre ± 0.035 BU plus native quantization allowance), unioned with all initial finite-difference offsets. All 133 state topologies and competitor exclusions are proved without shrinking the region after observing outcomes. Independently derived exact bases and all 16 coefficients agree for 1,251 scalar affine expressions. The full certificate costs 71.968 s and 49.160 MiB RSS. This justifies preparing a distinct reuse experiment with one continuous proof and per-state exact membership checks; The subsequent separately frozen full CPU pair preserves every coordinate and loss across 61 optimizer states, and all 150 final powers and predictions exactly. Complete worker costs are 393.110 s with the original point audits and 90.929 s with a freshly executed continuous proof plus 61 exact membership checks, an

observed ratio of 4.3233. The fresh continuous proof and final reconstruction are included. This single shared-machine pair has no timing confidence interval, energy measurement or universal-speed claim; the installed 0.1.2 add-on retains its original trainer.

4. Learning and generalization experiments

4.1 Iris geometry training and native roundtrip

The declared Iris split contains 120 training and 30 held-out examples. Training-only min/max preprocessing maps four features to four coherent sources plus a fixed coherent reference, with joint unit input power and no held-out clipping. Iris was previously examined in the project, so this experiment is not presented as a new blind discovery.

Historical learned delays were reset to an explicit untrained absolute geometry before initialization. Sixty fixed Adam updates decreased training loss from 6.2554405 to 0.3097067; training accuracy was 110/120 and held-out accuracy 27/30. All 61 represented states passed their geometry audits. The maximum initial loss-gradient finite-difference discrepancy was 1.4746713×10^{-6} against a 10^{-4} tolerance. Final fresh-geometry field reconstruction differed by at most $4.6194484 \times 10^{-16}$ against 10^{-11} . Full worker cost was 369.43 s, including 349.67 s of geometric auditing.

Actual Blender 4.5.14 applied the final native coordinates, saved a new file, reopened it with script execution disabled, recaptured geometry and rebuilt the own inference graph. All 150 observed powers had maximum difference zero and all predictions matched. The original .blend remained unchanged. This proves a bounded local native roundtrip, not external replication. A separately frozen secondary analysis independently audits the final native graph and certifies all 150 recorded fields, powers and class argmax decisions. Ideal train-only feature encoding from parsed binary64 CSV values is also enclosed; observed encoder error is at most 1.579×10^{-16} , field L1 error 2.7683×10^{-15} and power error 1.0366×10^{-15} within unchanged budgets. The minimum decision margin is at least 2.11197×10^{-4} . Of the 150 certified decisions, 137 are correct and 13 wrong: numerical stability does not imply class correctness. Input measurement/parsing uncertainty and intended/native-transform geometry errors remain unknown.

4.2 Prospectively fixed Wine comparison

Wine contains 178 examples and 13 measurements. The official original files and hashes are retained with attribution and licensing. Four features were fixed before training: alcohol, malic acid, ash and alkalinity of ash. One stratified split fixed 141 training and 37 held-out examples, and three initializations fixed seeds 1049, 1050 and 1051. The same training-only encoding, sources, class labels and split were used for all models. No held-out result selected an initialization, feature subset, step count or hyperparameter.

Linear softmax used 15 nominal coefficients; quadratic softmax used the 15 unique input monomials and 45 nominal coefficients. Both optimized a declared convex L2-regularized objective using fixed L-BFGS-B controls. Their function families differ from a passive optical network, so matching data does not establish equal physical constraints or capacity.

Model	Training correct	Held-out correct	Held-out Wilson 95% interval
Geometric, seed 1049	117/141	33/37	75.3–95.7%
Geometric, seed 1050	108/141	30/37	65.8–90.5%
Geometric, seed 1051	105/141	28/37	59.9–86.6%
Linear softmax	120/141	32/37	72.0–94.1%
Quadratic softmax	121/141	32/37	72.0–94.1%

All 183 geometry states and three final rebuilds passed. Maximum initial gradient discrepancy was 5.473×10^{-6} ; fresh field difference was at most 5.882×10^{-16} . The complete worker cost was 1114.17 s, including 1054.49 s of geometric audits. Baseline fits took 5.58 ms and 4.71 ms after imports, with imports included only in the total worker cost.

The three runs share the same 37 held-out observations and cannot be pooled as 111 independent trials. Paired exploratory exact McNemar p-values against either baseline were 1, 0.5 and 0.125, without multiple-comparison correction. No optical superiority is demonstrated. The spread across initializations shows sensitivity, and the single public-data split does not establish external blind or zero-shot generalization.

Linked figure: [All three Wine runs and matched comparisons.](#)

5. Actual NVIDIA execution and total-cost limits

The actual RTX 3090 was fixed by device UUID and admitted through the shared FIFO queue. Torch 2.6.0+cu124 with CUDA 12.4 and driver 581.29 evaluated complex128 fields, powers and the own 16-parameter Jacobians for 60 prepublished complex probes. Exact geometry and phase-argument preparation remained on CPU; this was not GPU triangle tracing. A separate functional autograd reference checked the weighted-loss derivative.

Maximum CUDA/NumPy differences were 3.61×10^{-16} in field, 3.89×10^{-16} in power, 5.55×10^{-14} in field Jacobian and 3.55×10^{-14} in power Jacobian; the autograd discrepancy was 1.56×10^{-13} . Independent rational certification of these 60 observed GPU probes gave field L1 bound $\leq 2.9356883 \times 10^{-15}$ and power bound $\leq 1.3807709 \times 10^{-15}$. Fifty-nine represented argmax decisions were certified; zero input remained unknown.

Batch	Cached CPU, ms	graph	Cached CUDA, ms	graph	Equivalent CPU, ms	dense	Equivalent CUDA, ms	dense
1	14.45		24.50		0.071		0.435	
150	17.43		27.06		0.499		0.559	
2048	75.00		25.72		8.597		0.897	

These are synchronized warm medians of five fixed repetitions for the same input/output contracts, including fields, powers and Jacobians. Preparation, transfers and complete worker costs are retained separately. The whole GPU benchmark worker cost was 6.72 s; peak RSS was 996.10 MiB and peak reserved CUDA memory 112 MiB.

Small graph batches are slower on CUDA, and the dense baseline is more efficient at all reported sizes. Warm batch scaling is not structural graph scaling, full native geometry-training speedup, host/device energy measurement or photonic-device efficiency. AMD hardware was not available and has not been executed.

A separately frozen full training pair executes the unchanged CPU trainer and own CUDA manual cross-entropy/Jacobian/Adam arithmetic for 60 updates, with 61 exact CPU geometry audits each. Both achieve 27/30 held-out predictions and identical final native-coordinate deltas; all 150 decisions agree, maximum power difference 3.8858×10^{-16} . Across CUDA states, maximum matched CPU field/power/loss/gradient differences are 5.404×10^{-16} , 6.662×10^{-16} , 8.882×10^{-16} and 2.843×10^{-13} . Fresh final geometry agrees within 5.118×10^{-16} . Full-worker CPU/CUDA times are 370.46/390.09 s; this pair provides no global speedup. Geometry audits dominate at 350.49/363.40 s. CUDA loss/gradient/proposal time is 2.12 s, versus 1.17 s for the CPU training updates, with phase preparation/transfers and CPU controls also retained. GPU reserved memory peaks at 22 MiB.

This is one pair, not a timing confidence interval; exact geometry and native quantization remain CPU.

A subsequently frozen native graphics frontier constructs the network directly from the five source rays, with new reflection/transmission queries rendered against the 6,656 evaluated triangles in 17 batches. Three actual RTX3090 native OpenGL constructions each exactly verify all 133 graphics surface/primitive candidates without CPU replacement, producing 133 states, 186 edges and 17,060 represented terminal paths. Exact nodes match fresh CPU and prior native geometry; an independent full graph audit passes. All 150 phase-aware outputs preserve predictions, with maximum field/power differences $5.1179 \times 10^{-16}/5.5512 \times 10^{-16}$. Coherent phase arithmetic uses CPU exact refined lengths rather than graphical depth. Full fresh CPU construction costs are 30.7660/30.4246/29.8658 s versus 31.5350/32.1434/33.0084 s for GPU graphics plus exact verification: no full construction speedup is observed. Candidate verification dominates the latter at 30.5500/31.1753/32.0184 s. This is actual graphical geometry execution, not matrix multiplication substituted for a scene, but it does not imply hardware RT cores, physical fidelity or novel graphics tracing. A separate frozen close-plane diagnostic selected the wrong plane at a 2^{-25} BU spacing; that failure is retained and motivates exact verification.

A separately frozen optimizer recovery trial actually terminates only an owned research worker after its atomic checkpoint commits 12 updates. A fresh process repeats the whole original-family proof and deterministically recomputes the 12-update prefix before accepting coordinates and Adam moments. All 61 coordinate/loss states and all 150 final powers/predictions equal the uninterrupted reference exactly; both obtain 27/30 held-out correct. Four actual checkpoint adversaries are rejected, including forged optimizer moments with a recomputed self-consistent hash. Original checkpoint bytes are preserved. Full uninterrupted/interrupted/resumed costs are 96.8763/75.5020/94.5633 s, total 267.2220 s and peak owned RSS 56.59 MiB. This validates process-level research-worker recovery, not yet the installed Blender resume UI, whole-host crashes or power-loss durability. The first frozen attempt failed in final metadata before the interruption stage; that null outcome is retained.

A prospective second native graphics profile replaces the expensive candidate fan-support routine with exact plane intersection, independent Gram closed-triangle predicates and a positive square-perimeter interior witness, followed by unchanged exhaustive independent nearest-object exclusion. Three original CPU builds cost 29.8096/30.0367/29.9906 s versus 10.5964/10.5326/10.4500 s for GPU graphics plus the new exact refinement, with all gates, 133 states, 17,060 paths and 150 outputs preserved. This approximately 2.85 median-ratio observation is a combined algorithmic-route improvement, not an isolated GPU benefit: that observation alone leaves the identical-refinement candidate-engine ablation unresolved; the subsequent study below tests it. Three shared-host repetitions with fixed ordering do not establish a general speed factor. Full process cost is 134.7320 s and peak owned RSS 348.40 MiB.

A subsequent prospectively fixed alternating three-pair ablation uses native Blender CPU BVHs and GPU raster depth candidates with identical exact selected-plane, Gram, interior and exhaustive-nearest refinement. All six graphs and all 150 outputs pass. Full CPU construction costs 9.897551/9.829980/9.868582 s versus 9.964846/10.004324/9.912861 s for GPU construction (median CPU/GPU ratio 0.99034). No global GPU advantage is observed in this scene. Proof dominates both routes at approximately 9 s. The full supervised process costs 72.844940 s with peak owned RSS 364.90 MiB. This controls the earlier combined improvement and precludes attributing its 2.85 ratio to GPU hardware. Three alternating exploratory pairs do not establish universal scaling or energy benefits.

A further separately frozen trial executes coherent transport in actual native OpenGL fragment shaders. The shader receives captured planes/materials, current geometric rays, wavelength and incoming complex fields. Raster/depth selects each surface; FP64 arithmetic derives optical distance, phase, splitting, mirror phase and terminal-reference phase. No expected hit IDs, precomputed transfer matrix or edge phase weights supply that computation. Exact CPU topology admission and compensated coherent merging remain explicit. Three 150-input repetitions each make 19,950

actual geometric queries and preserve every native prediction, with maximum complex-field discrepancy 7.5011×10^{-14} and power discrepancy 7.2165×10^{-15} , below unchanged 1×10^{-11} gates. Four actual GPU phase/interference controls pass; coherent versus intensity-sum power differs by up to 0.27376. Transport plus CPU merging costs 2.5075/3.2332/3.4803 s; full supervised cost is 36.6185 s and peak owned RSS 403.04 MiB. CPU phase-only graph reuse is a different workload and supplies no equivalent speedup claim. Ten preceding null trials preserve interface, integer-readback and shader-lifecycle failures. The successful trial verifies all binary64 input echoes and rejects every newly raised GL error. GPU gradients, rigorous bounds on unobserved GPU rounding, physical fidelity, AMD and superiority remain unestablished. [Protocol and retained evidence](#).

Linked figure: [Captured-surface coherent fragment transport](#).

A targeted additional primary-source review establishes relevant graphics/optical precedents: [Purcell et al. 2002](#) maps ray tracing to programmable graphics stages; [MacroSim institutional documentation](#) describes double-precision GPU/OptiX optical tracing and a 3D scene editor; [Mauch et al. 2014](#) explicitly includes coherent tracing in its publisher abstract; [Keksel et al. 2023](#) models optical instruments with configurable OptiX laws and compares virtual images with its physical focus-variation instrument. These precedents rule out treating graphics-engine optics or coherent GPU tracing alone as our original contribution. No implementation was integrated or reproduced. The full 2014 article was inaccessible; the reviewed 2023 method passages do not establish equivalence to this network. Primary Crossref metadata confirms the 2013 MacroSim DOI 10.1117/1.OE.52.5.053004, correcting the mismatched institutional link. [Review and source-access receipt](#). The historical literature extraction backlog remains unchanged and novelty unestablished.

A separately fixed independent rational secondary analysis audits the native geometry and bounds all 450 observed graphical outputs, after exact equality of the three repeated field/power arrays is checked. Maximum field L1 error is bounded by 9.9736×10^{-14} for recorded binary64 inputs and 9.9745×10^{-14} for ideal train-only encoding; corresponding power bounds are 7.1507×10^{-15} and 7.0918×10^{-15} . All four phase/coherence controls pass their interval error budgets. All 150 unique argmax decisions are certified, including 137 correct and 13 wrong predictions; repeated runs do not add independent biological observations. The minimum certified margin is at least 0.00021119712. Full analysis costs 21.0174 s with peak owned RSS 105.49 MiB. These certificates enclose the observed GPU arithmetic and CPU merges against the exact represented model, not every unobserved GPU execution or physical optics. [Full interval evidence](#).

Native captured-surface graphical derivatives

A separately frozen native graphics trial differentiates all 16 paired-mirror X translations for five fixed probes (three native Iris rows, one bounded complex input and zero). FP64 fragment shaders derive intersection-distance, phase and complex-field tangents from captured planes/materials and current incoming value/tangent. Exact CPU affine origin jets, topology admission, compensated coherent merges and the power/loss chain rule are explicit. No transfer or optical Jacobian matrix is fed to the shader. All gates pass: field discrepancy 6.5773×10^{-14} , field-Jacobian discrepancy 5.7353×10^{-12} , power-Jacobian discrepancy 9.2771×10^{-13} and diagnostic loss-gradient discrepancy 5.4321×10^{-12} . Thirty-two independently audited materialized perturbed geometries give central finite-difference scaled discrepancies $7.4685 \times 10^{-10}/1.2831 \times 10^{-9}/2.6588 \times 10^{-9}$ for field/power/loss. Zero-input derivatives are exactly zero. The 16 passes execute 21,280 geometric queries; full supervised cost 336.9243 s and peak owned RSS 434.86 MiB include 296.7458 s of independent finite-difference audits. All 8,084 owned GL stages and exact input echoes pass. This validates observed hybrid graphical derivatives for the stated probes; it does not demonstrate full native graphics training or universally enclose derivatives. [Protocol and raw evidence](#).

6. Separate wave-reference experiment

An original coherent complex128 angular-spectrum solver propagates a declared Gaussian boundary field in a homogeneous outgoing scalar Helmholtz half-space. Its continuum reference is independently implemented Fourier–Bessel/Hankel quadrature with analytic Gaussian spectrum and separate quadrature refinement. A paraxial Gaussian solution is an additional approximation reference. The transverse integral of $|E|^2$ in a circular aperture is the declared observable; it is not a vector Poynting flux or a captured detector’s modal-coupling measurement.

The fixed family uses waist 0.1 BU, aperture radius 0.15 BU, wavelengths 0.1/0.025/0.00625 BU and propagation distances 1/2/4 BU. The first 36 observations completed, but the $\lambda=0.1$ BU, $z=4$ BU case failed the fixed 2×10^{-6} radial-field tolerance at window 16 BU: difference 2.2311874×10^{-6} . Refining $512^2/1024^2/2048^2$ at that window left the same error. This negative result was published before any remedial trial.

A separately frozen adaptive family used windows 32/64 BU and grids $1024^2/2048^2/4096^2$, with unchanged scientific tolerances and a larger declared RAM envelope. All nine physical cases passed their prefixed gates. Maximum radial-field differences were 4.0866539×10^{-7} at window 32 and 9.2920364×10^{-8} at window 64, in the 65 checked points. Aperture integration remained sensitive to spatial resolution; the coarse $1024^2/32$ mesh had maximum fraction error 0.00289647, retained rather than called precise. Fine and expanded meshes satisfied their own gates. Complete remedial cost was 133.05 s, peak RSS 1866.79 MiB.

Quadrature refinement is numerical convergence evidence, not a rigorous universal FFT error bound. This Gaussian mode was not inferred from the network capture. The separate study therefore characterizes a declared propagation regime without establishing full-network physical fidelity or providing a correction factor for its detector powers.

Linked figure: [Retained negative and separately frozen remedial wave family](#).

7. Installation, replication and artefact status

The standalone own Blender addon packages the original solver dependency closure, semantic contracts, own trained example, dataset, source provenance and integrity manifest. Its UI uses explicit coherent inputs, background Blender jobs, owned cancellation, stale-result rejection, completed-result recovery and new-file saving. Original implementations and immutable historical ZIPs are retained.

Two native attempts are retained with null metrics: a version-string/preflight failure before installation, and a successful isolated installation followed by an incorrect complex serialization handoff. ZIP 0.1.1 corrects that handoff with the already used coherent wire converter. Its third native audit passed all nine predefined controls in 729.01 s with peak aggregate RSS 573.49 MiB. Own training actually ran inside background Blender, audited all 61 states and reproduced the previous 27/30 held-out predictions, loss trajectory and all 150 powers with observed difference zero. Stale inputs/geometry, real cancellation, completed-job recovery, atomic apply, copy/reopen and modified-result rejection also passed. A subsequent shutdown log contains a double-unregister RuntimeError after the audit explicitly unregistered a still-enabled addon. It does not change the nine numerical/control outcomes, and it motivated a separately frozen lifecycle regression. ZIP 0.1.2 changes only lifecycle code, installation text and manifest; all scientific-worker, vendor and data bytes remain identical to 0.1.1. Five actual native lifecycle controls pass in 65.04 s with peak aggregate RSS 516.71 MiB, including duplicate registration, Blender utility cycles, owned-process cancellation on disable and native150 inference parity. The entire post-exit log contains no traceback or unregister error. The original failure remains archived. Host-crash recovery and human usability remain open.

Every bounded scientific record includes a published profile, source hashes, raw outcomes and evidence index. Reproducing old protocols requires their corresponding commit or archived preimages; current source evolution must not be substituted silently. A clean local installation is distinct from an execution in an independent environment, and both are distinct from a replication performed and interpreted by independent researchers. A separately frozen Linux CPU run on GitHub, published at commit `bbc06c3ace138d98873df9ba0a4d83252ebfetc8`, independently checks 40 source/profile/receipt blobs before science. All 34 selected controls, certificate replay and 61 training-state audits pass. It reproduces 27/30 held-out predictions and all 150 decisions, with maximum power difference 3.8858×10^{-16} ; scientific-worker time is 367.37 s and peak aggregate RSS 88.29 MiB. Setup/upload are separate job stages. This is external-environment reproduction of our code, not independent human replication or GPU execution. The original YAML setup failure is retained. A separate native Linux protocol published at `c1642ca090e8ba40bcd64f875b0d9ec39b4208ae` downloads and verifies the official Blender archive and executable before isolated ZIP 0.1.2 installation. All nine native controls, 61 training-state audits and the post-exit lifecycle log pass in actual Blender 4.5.14 LTS on Linux x86_64, Python 3.11.15 and NumPy 1.26.4. It achieves 27/30 held-out predictions; power differences are 3.8858×10^{-16} versus Windows and zero versus its final native geometry. Full controller cost is 685.77 s (21.09 s preparation, 663.69 s native controls), peak aggregate RSS 756.34 MiB. The first HTTP403 preparation failure is retained. This is external native software reproduction, not independent expert interpretation. The latter human replication has not been obtained. No specialist critique, journal submission, external registration receipt or physical experiment is asserted.

8. Related work and novelty boundary

Trainable coherent photonic networks and diffractive optical learning predate this implementation. [Shen et al.](#) and [Lin et al.](#) establish that context. [Kawata and Hirose](#) is an earlier coherent path-difference learning precedent. These works prevent presenting optical learning itself as our novelty.

[Ho et al.](#) combines ray and diffraction-based optical modelling with optimization and coherent demonstrations, an important closest-method comparison. Its image-formation conventions and sampling scope cannot simply replace the common-coherence multi-input readout here. [Chromatix](#) provides a separate differentiable wave-optics precedent. We have not installed or benchmarked it in this study.

[DeepG](#) and [QA-IBP](#) show related certification precedents for geometric transformations and quantized inference. Their reviewed scopes do not establish exact optical triangle-path equivalence, but they prevent broad claims that combining geometry, decisions or actual arithmetic certification is automatically unprecedented. [Matsushima and Shimobaba](#) is a known angular-spectrum sampling-method reference; this work does not claim to have implemented its band-limited method or completed a full-text review of that paper. [Steck's optics notes](#) support the declared Gaussian conventions.

[Knoll et al.](#) establishes interval/affine inclusion for ray tracing. [de Koning et al.](#) optimizes optical geometry with differentiable non-sequential rays and reports LightTools checks. [Chen et al.](#), [RFDT](#) is a close coherent multipath/geometry-gradient precedent with edge-diffraction transitions. Those transitions differ from our hard-branch admissibility boxes; equivalence is not assumed. A [bounded primary review](#) records inspected passages and file hashes. These three supplementary identities have not been merged or deduplicated into the historical counts below.

The current literature register has 194 raw entries, 175 unique identities and 30 completed extraction records; 106 requested extractions remain pending. These counts are historical audit status, not an exhaustive absence-of-prior-art proof. The implemented combination is original code and an integration contribution. Exceptional scientific originality, exceptional importance and durable

impact have not been demonstrated. A defensible novelty claim requires closer matched comparisons and independent expert assessment.

9. Conclusions and open tests

The bounded evidence supports an own geometry-driven coherent scalar classifier, exact admitted branch coverage, useful certificates of observed represented outputs, own gradients/training, local native Blender roundtrip and actual NVIDIA evaluation. It also exposes meaningful limitations: optical comparisons do not show superiority, small CUDA batches are slower, finite FFT windows can fail, and the current passive readout has restricted quadratic capacity.

The remaining research programme includes bounded input/parameter uncertainty and native-transform error budgets, bounded trained-batch uncertainty beyond the completed arithmetic certificate, broader geometry/task families, structural scaling and ablations, full-cost and energy measurements, actual AMD execution, completed standalone acceptance, broader independent-environment reproduction beyond the completed native Linux Blender trial, outside expert replication, full-network physical modelling and calibrated measurements when physical-device claims are sought. None is replaced by a prize aspiration or by a passing software test.

Data, code and reproduction map

Repository: [Agnuxol/Neuro3D](#), main with scientific history preserved; [PR history 6](#). Software license: MIT; Wine data attribution/license: [UCI Wine](#), DOI [10.24432/C5PC7J](#). No third-party optical implementation is bundled in the own addon.

Evidence	Protocol / method	Raw record
Represented output intervals	method and limitations	analysis02 index
Observed native batch and ideal encoder	150-row certificate	certificate
Conditional continuous geometry boxes	fixed protocol	four boxes, unknowns retained
Own Iris training	frozen protocol	training result
Actual native save/reopen	native reproduction	native result
Wine and matched baselines	comparison	complete result
Actual CUDA training and paired full cost	training protocol	CPU/CUDA pair
Complete operator rank and normalization	Gram protocol	rational operator certificate
Whole continuous training family	family protocol	exact affine/topology proof
Fresh continuous-proof training full cost	paired protocol	full equivalent pair
Identical-proof native CPU/GPU geometry	prospective ablation	six full builds
Captured-surface coherent fragment fields	protocol and ten null trials	actual successful GPU trial
Observed graphics output and decision intervals	independent certificate	rational enclosure
Captured-surface derivatives	frozen probe protocol	gradients and 32 geometric finite differences
Actual NVIDIA	benchmark	indexed evidence
Gaussian negative/remedial	original, remedial	original result, remedial result
Standalone addon acceptance	versioned protocol, failures and lifecycle limitation	nine-control native result

External native Linux Blender	fixed version, download failure and completed run	native external result
External Linux CPU reproduction	profile, failed setup and completed run	actual external result

Publication commits fixed before experiments are recorded in each linked protocol. Hashes and preimages must be checked before interpreting or rerunning a result. External expert review and author approval of a submission-ready manuscript remain pending.